

A discrete kinetic scheme for simulating anisotropic liquid-solid phase transition with thermal convection



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Abstract

Dendrite growth is a complex process involving interface dynamics and material thermodynamics. A discrete kinetic scheme, derived from a modified Boltzmann equation^[1], is proposed to quantitatively evaluate the liquid-solid phase transition with anisotropic interface energy. Utilizing finite-volume discretization and numerical solutions along characteristic lines, this scheme offers excellent stability and accuracy, and is robust on non-structured meshes. Additionally, it is coupled with the Boltzmann-Shakhov equation, which is solved using the discrete unified gas kinetic scheme (DUGKS)^[2], to model phase transition under various hydrodynamic conditions through thermal effects and diffuse interface-based fluid-solid interactions. Simulations results imply that the proposed scheme is highly effective in capturing the motion of complex interfaces, considering the thermal and hydrodynamic interactions between the two phases.

References

[1] D. Sun, *Appl. Math. Lett.* **103**, 106222 (2020).

[2] Z. Guo, R. Wang, K. Xu, *Phys. Rev. E* **91**, 033313 (2015).

Methods

The evolution equation for **anisotropic dendrite growth**

$$a_p \frac{\partial \beta}{\partial t} + \xi \cdot \nabla \beta = -\frac{1}{\tau_c} (\beta - \beta^{\text{eq}}) + S_p,$$

where

$$\beta^{\text{eq}} = \frac{\phi}{(2\pi R_p T_p)^{D/2}} \exp \left[-\frac{(\xi - \mathbf{u}_p)^2}{2R_p T_p} \right].$$

The scheme for time-advancing step is

$$a_p (\beta_k^{n+1} - \beta_k^n) + \frac{\Delta t}{V_k} \mathcal{F}_k^{n+1/2} = \frac{\Delta t}{2} (\Omega_{p,k}^n + \Omega_{p,k}^{n+1}) + \Delta t S_{p,k}^n,$$

and the scheme for flux reconstruction step

$$\begin{aligned} a_p \beta(\mathbf{x}_b, t+s) &= \beta(\mathbf{x}_b - \xi s, t) + (a_p - 1) \beta(\mathbf{x}_b, t) \\ &+ \frac{s}{2} [\Omega_p(\mathbf{x}_b, t+s) + \Omega_p(\mathbf{x}_b - \xi s, t)] \\ &+ s S_p(\mathbf{x}_b - \xi s, t), \end{aligned}$$

where the numerical solution along its characteristic lines are used, and the operator splitting method was applied to address the anisotropic function.

The evolution equation for **thermal flow**

$$\partial_t f + \xi \cdot \nabla f + \mathbf{a} \cdot \nabla_\xi f = \Omega_f,$$

where the collision operator is approximated by Shakhov model, and the equation is solved by the DUGKS method.

Results

Dendrite growth driven by heat diffusion

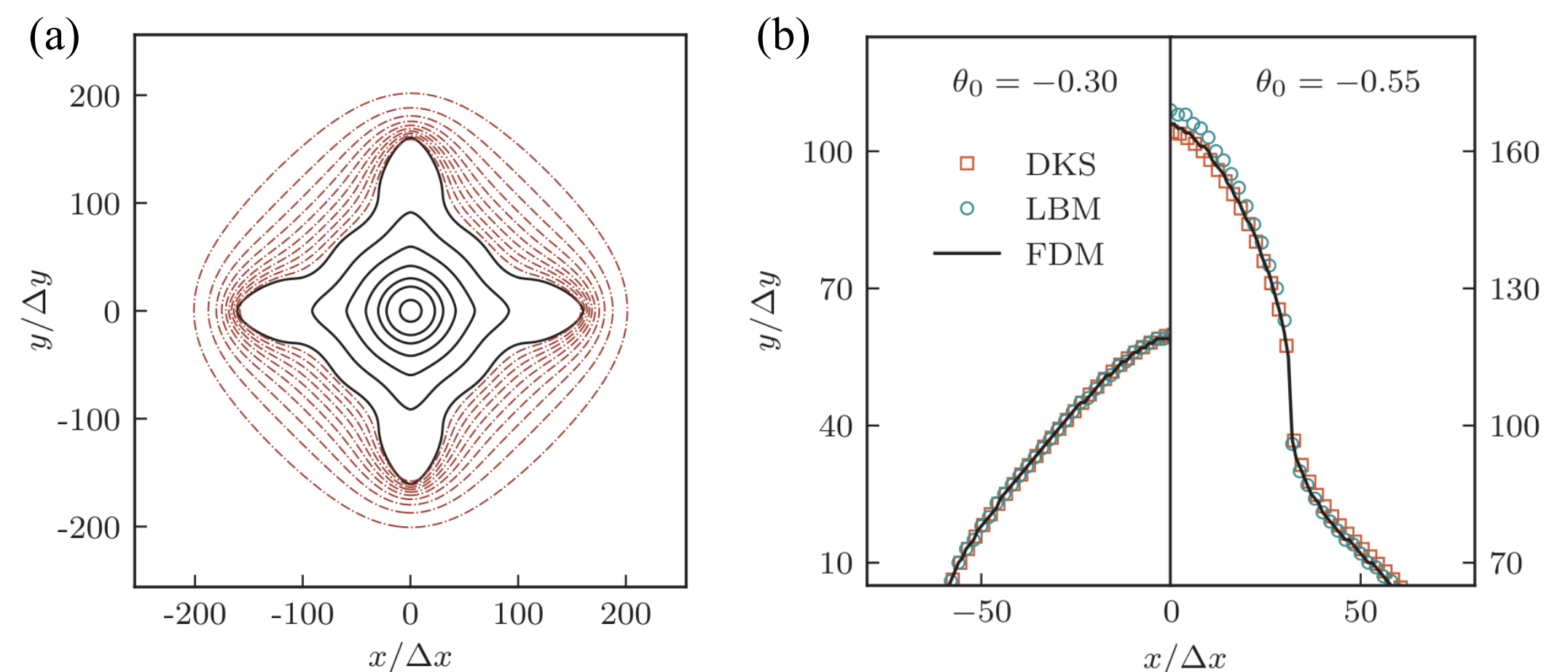


Fig 1. (a) The history of equiaxed dendrite morphology at $t/\tau_0 = 0, 4, 8, 16, 32, 64$ and 128 , and isothermal contours at $t/\tau_0 = 128$, (b) tip curves by the DKS, LBM and FDM.

Dendrite growth under forced convection

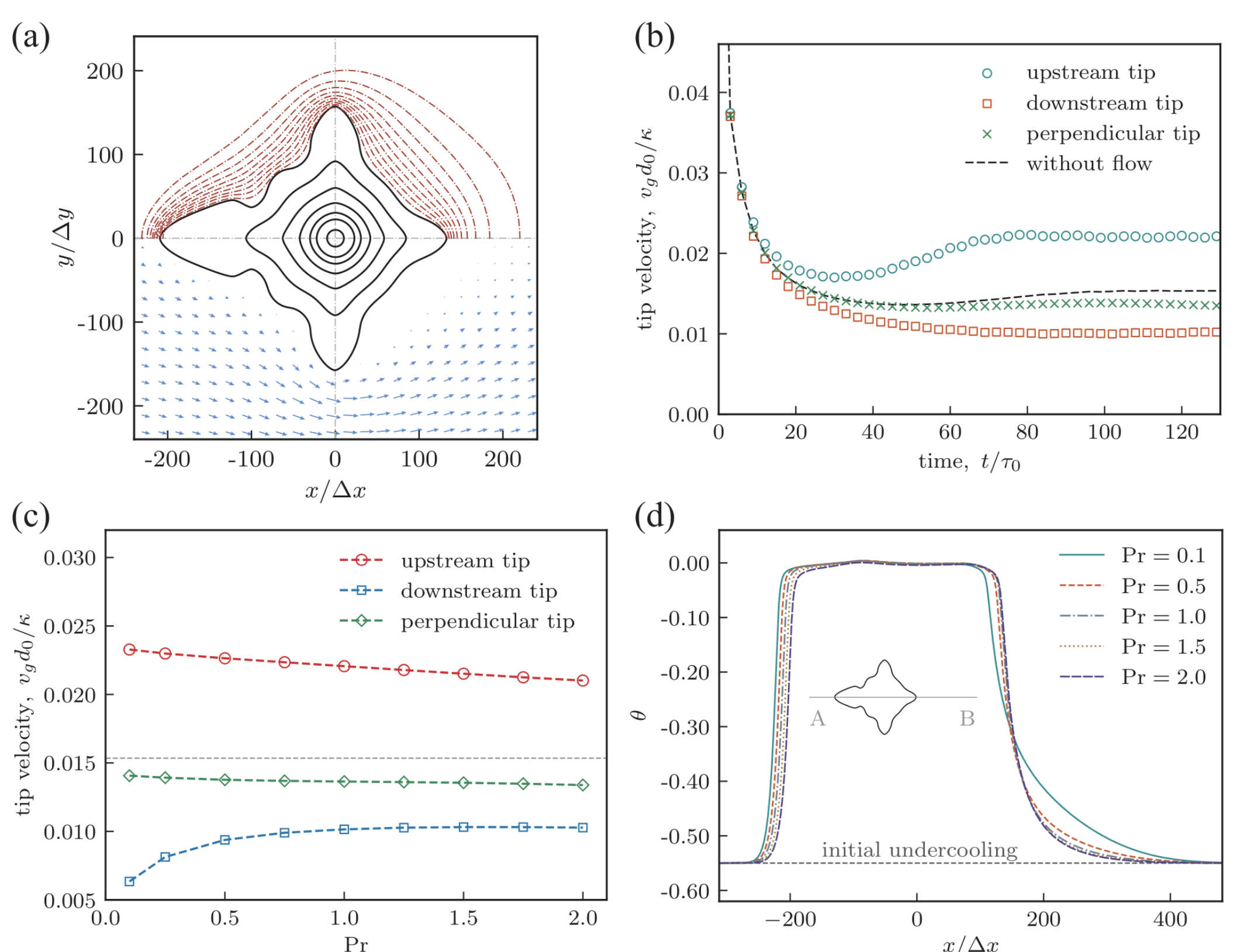


Fig 1. (a) The equiaxed dendrite morphology and flow streamlines, (b) growth velocities of upstream, downstream and perpendicular tips, (c) steady velocities of tips at different Prandtl numbers, (d) undercooling profiles along horizontal axis at time $t = 128\tau_0$.

Conclusions

- 1 A novel discrete kinetic scheme evaluating the process of anisotropic phase transition was proposed.
- 2 Coupled with DUGKS solving hydrodynamics, the model has the capability to capture phase change thermal effects and fluid-solid interaction around the dendrite interface.
- 3 This study implies a significant potential of kinetic theory in modelling materials phase change dynamics interacted with thermal flows.